Discussion draft [do not cite or distribute] November 2026

Online Appendix

The Uneven Reach of Accountability: Verification and Public Goods in China

Aodi Cheng*

November 2026

Tables, figures, equations, hypotheses and sections numbered without a letter refer to the paper.

Contents

- Appendix A: Sample and Measurement 1
- Appendix B: Additional Estimates and Robustness Checks 3
- Appendix C: Text Analysis 11
- Appendix D: Further Evidence on the Mechanism 16

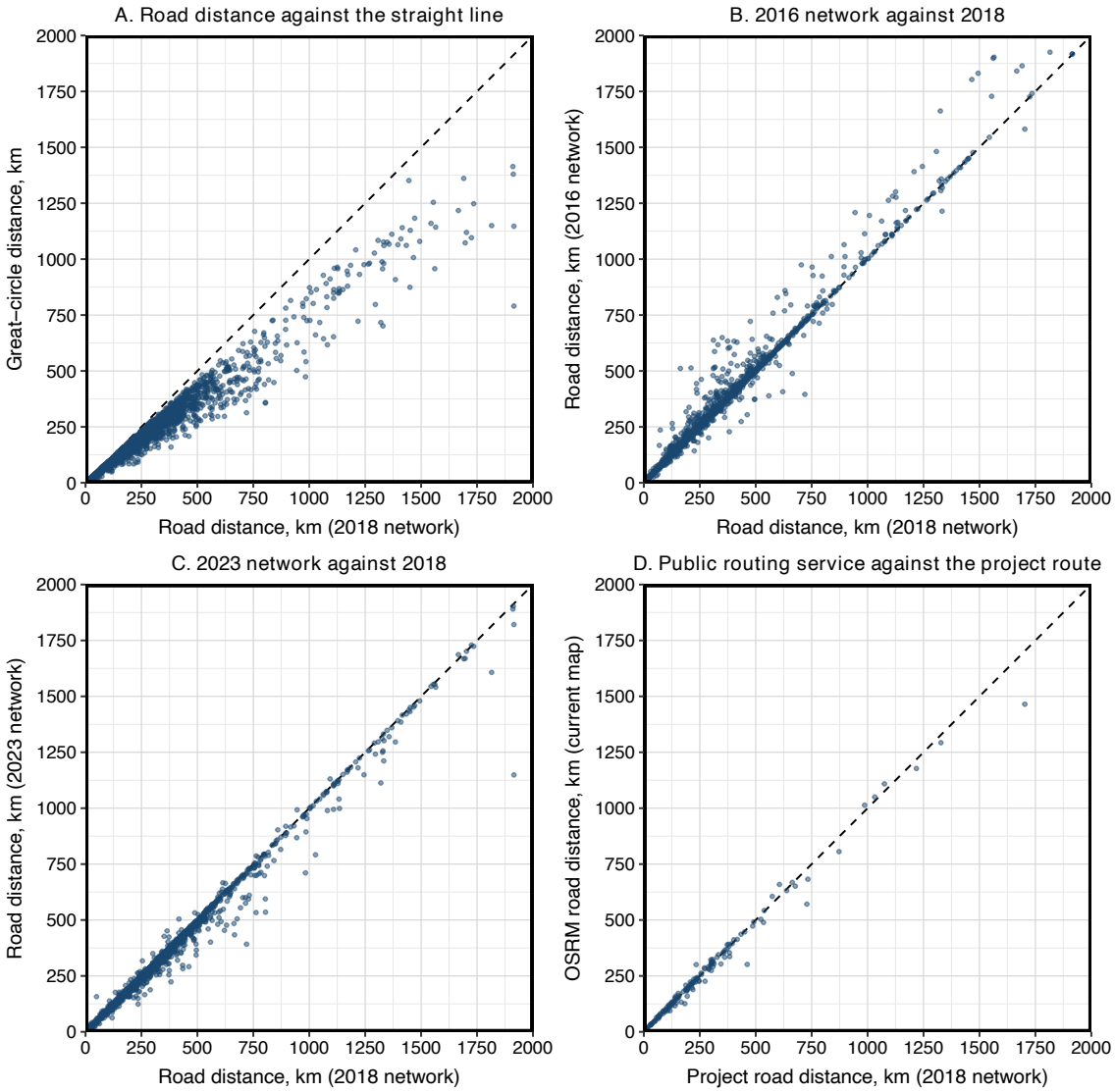
A Sample and Measurement

Figure A1 plots the four checks on the road distance summarised in Section 5.2: the road distance against the straight line, the 2016 and the 2023 networks against the 2018 network, and a public routing service against the project route. Figure A2 shows the distribution of road distance and of travel time in the estimation sample, with counties grouped by quarter of within-province remoteness.

*University of Minnesota, Twin Cities. Email: che00319@umn.edu.

Discussion draft [do not cite or distribute] November 2026

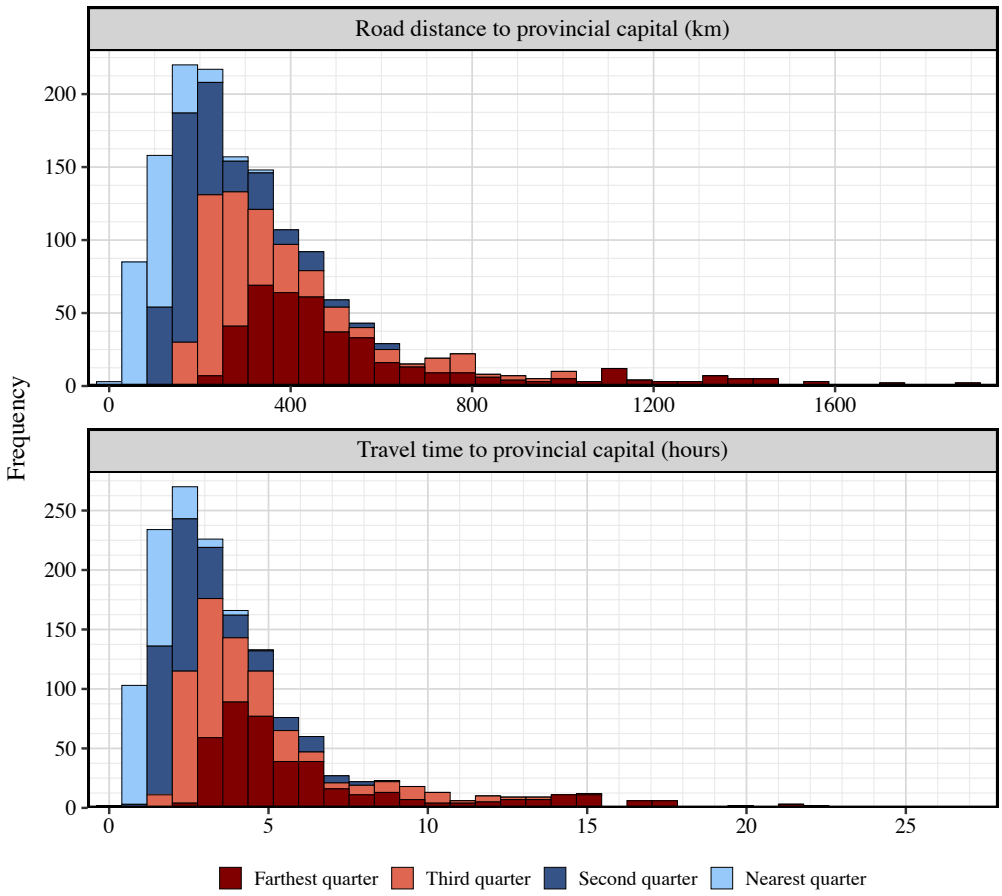
Figure A1: Checks on the road distance to the provincial capital.



Notes: Each point is a county; the dashed line is the 45-degree line. A: the road distance of equation (3) on the 1 January 2018 OpenStreetMap major-road network against the great-circle distance from the county centroid to the capital, 2,676 counties (correlation 0.98). B and C: the road distance on the 1 January 2016 and 1 January 2023 networks against the 2018 network (correlations 0.99 and 0.99; median ratios 1.00 and 0.99). D: the distance returned by the OSRM public routing service on the current OpenStreetMap network, all road classes, against the project route, for 155 counties drawn five per province by quintile of road distance (correlation 0.99; median ratio 1.01).

Discussion draft [do not cite or distribute] November 2026

Figure A2: Road distance and travel time to the provincial capital on the 2018 network.



Notes: Estimation sample, 1,456 county towns in 27 provinces. Upper panel: shortest route on the OpenStreetMap major-road network of 1 January 2018 from the county centroid to the provincial capital. Lower panel: travel time along that route at class-specific speeds. Counties are grouped by quarter of the within-province percentile of road distance, the remoteness measure R_i; the farthest quarter of a small coastal province lies closer to its capital in kilometres than the nearest quarter of a western one.

B Additional Estimates and Robustness Checks

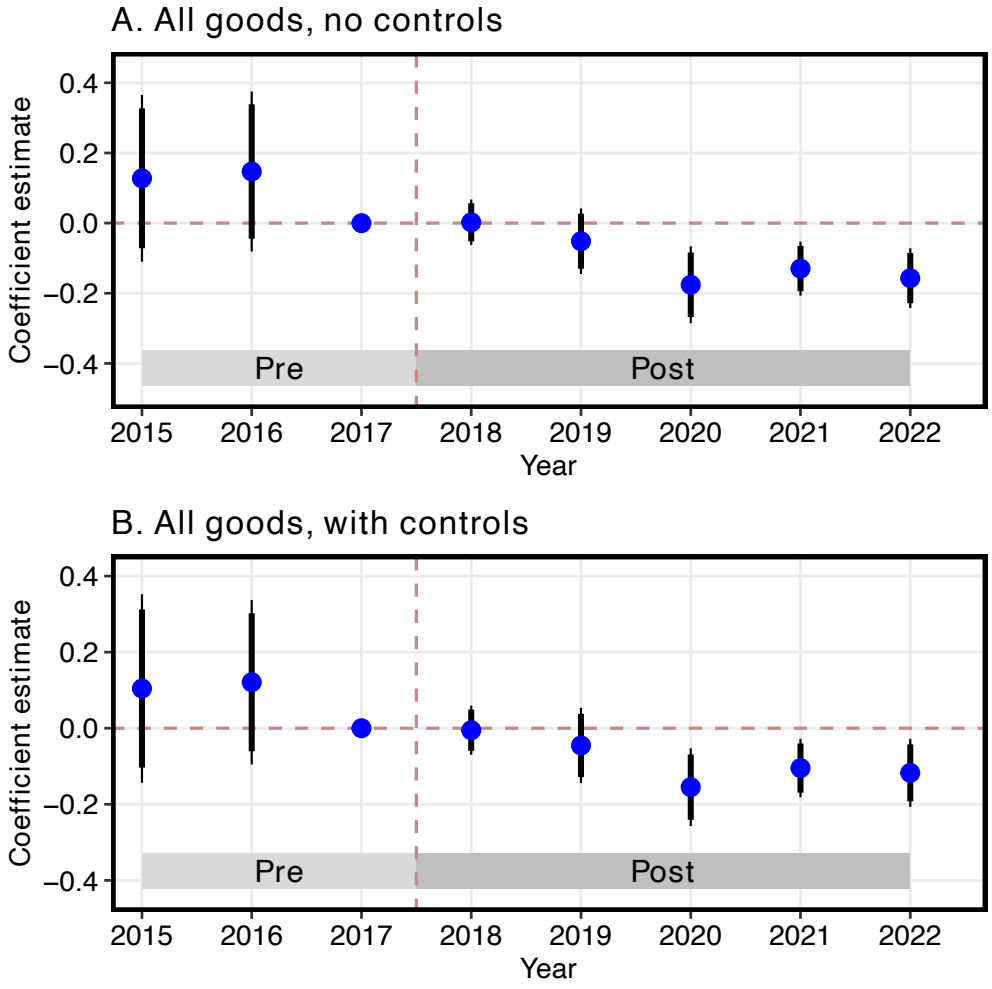
This appendix reports the estimates that the main text summarises: the event study for the all-goods index, the placebo distances, the two inference procedures, the sensitivity of the estimate to measurement choices, the main specification with proportional quantities and with the binary pressure measure, the checks on the composition of the sample, the split of the index by goods, and the event studies that hold the deep-poverty areas fixed.

B.1 Event Study for the All-Goods Index

Figure B1 reports the event-study specification of equation (9) for the all-goods index, without the controls in panel A and with them in panel B; Figure 4 in the main text reports the two family indices. The two pre-period coefficients are positive and jointly indistinguishable from zero, the coefficients for 2018 and 2019 are close to zero, and the coefficients from 2020 onward are negative with 95 percent intervals that exclude zero. Adding the controls leaves the path unchanged.

Discussion draft [do not cite or distribute] November 2026

Figure B1: Event study for the all-goods index: the pressure-specific remote–accessible gradient relative to 2017.



Notes: Coefficients $\hat{\mu}_k$ on year × Remote × Discipline from equation (9), all-goods index, reference year 2017; county and province × year fixed effects. Panel A has no controls; panel B adds the controls of Table 4. Thick bars are 90% and thin bars 95% intervals from province-clustered standard errors. In panel A the pre-period coefficients are 0.13 (2015) and 0.15 (2016) and the post-period coefficients 0.00 (2018), −0.05 (2019), −0.18 (2020), −0.13 (2021) and −0.16 (2022); in panel B they are 0.10 and 0.12, then −0.01, −0.05, −0.15, −0.10 and −0.12. The joint test of the 2015 and 2016 coefficients gives F = 0.80 (p = 0.45) in panel A and F = 0.62 (p = 0.54) in panel B; the average post-period coefficient is −0.10 in panel A and −0.09 in panel B.

B.2 Placebo Distances

Hypothesis 2 holds that the gradient belongs to the provincial capital because the capital is where the apparatus that verifies a county's delivery is based. Remote county towns are also far from the coast, from markets and from the national centre, and a gradient driven by economic remoteness would look the same in Table 4. The placebo design of Mastrorocco and Teso (2026) separates the two readings: a distance to a city that is not the county's principal should carry no gradient of its own and should not absorb the gradient to the capital. I compute the within-province percentile of great-circle distance from each county to Shanghai, Guangzhou and Hangzhou, the three economic centres of the coast, and to Beijing, and add the placebo percentile with its two interactions to equation (6), dropping the province whose capital is the placebo city. The within-province percentile is the appropriate form, because the design compares counties within provinces and because the rank of a county by

Discussion draft [do not cite or distribute] November 2026

distance to Shanghai within, say, Sichuan is a well-defined and quite different ordering from its rank by distance to Chengdu.

Table B1 reports the result. The capital gradient keeps its size in every column, without and with the controls, against the baseline of Table 4. The economic-centre gradients are positive in every column, and the Guangzhou gradient is positive and statistically significant: holding distance to the capital fixed, county towns far from the economic centre of the south did relatively better under pressure after 2018. The sign runs against an economic-remoteness account, under which county towns far from the coast should have fallen behind. Beijing is the seat of the central principal and is reported for completeness; its gradient is positive and insignificant, and the capital gradient is unchanged beside it, so that the relevant point of reference is the provincial capital. The three economic-centre placebos pass the test that Hypothesis 2 sets.

Table B1: Placebo distances: the gradient belongs to the provincial capital.

	Shanghai		Guangzhou		Hangzhou		Beijing	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post × Remote × Discipline	−0.201**	−0.168**	−0.186**	−0.152*	−0.200**	−0.166**	−0.213**	−0.179**
	(0.080)	(0.080)	(0.078)	(0.078)	(0.079)	(0.079)	(0.081)	(0.082)
Post × City × Discipline	0.121	0.114	0.206***	0.196**	0.135	0.127	0.107	0.101
	(0.084)	(0.086)	(0.073)	(0.073)	(0.082)	(0.084)	(0.080)	(0.083)
Post × Remote	0.049	0.017	0.077	0.045	0.058	0.025	0.064	0.032
	(0.072)	(0.073)	(0.067)	(0.068)	(0.073)	(0.074)	(0.068)	(0.068)
Post × City	0.012	0.018	−0.013	−0.006	0.008	0.013	−0.030	−0.026
	(0.074)	(0.076)	(0.060)	(0.061)	(0.072)	(0.074)	(0.072)	(0.072)
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Province × Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls		✓		✓		✓		✓
Provinces	27	27	26	26	26	26	27	27
Counties	1,456	1,454	1,419	1,417	1,424	1,422	1,456	1,454
Observations	11,514	11,483	11,219	11,188	11,258	11,227	11,514	11,483

Notes: All-goods index, equation (6). Each pair of columns adds the within-province percentile of great-circle distance to the named city (City), interacted with Post and with Post × Discipline, alongside the capital terms, and drops the province whose capital is that city; Shanghai and Beijing have no county towns in the sample, so columns (1), (2), (7) and (8) keep all 27 provinces. Odd columns are without controls and even columns add the controls of Table 4, whose columns (1) and (2) report the specification without a placebo distance. Standard errors clustered by province in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.3 Randomisation Inference and the Wild Cluster Bootstrap

One potential issue is that the variation in the moderator is concentrated at the provincial level. Although the dependent variable is observed at the county-year level, the effective variation in D_p comes from 27 provinces, and conventional cluster-robust standard errors may understate the uncertainty. Every estimate in this paper is therefore accompanied by two procedures that do not rely on asymptotics in the number of clusters. They test different things, and the difference matters for reading the tables.

Randomisation inference targets the provincial moderator (Ho and Imai 2006). The logic of the test is straightforward: the coefficient β_2 depends on the alignment between the post-2018 change in each province's remote–accessible gradient and that province's pressure value, and the permutation test asks whether this alignment is stronger than the alignment that arises when the same 27 pressure values are reassigned at random across the 27 provinces. I permute the pressure values 999 times and re-estimate β_2 under each assignment, keeping outcomes, remoteness, controls, fixed effects and the sample unchanged; the Fisher two-sided p-value is the share of placebo

Discussion draft [do not cite or distribute] November 2026

estimates at least as large in absolute value as the observed one. The placebo coefficients are centred on zero, and the observed estimate lies in the lower tail of both distributions, so that only a small fraction of random assignments of pressure produce a gradient as large as the observed one. The test treats the 27 pressure values as exchangeable across provinces under the null: it asks how unusual the observed pairing of pressure and gradient change is among all pairings of the same values. Pressure was set by the course of the campaign and by each province's history, so that exchangeability is an assumption the test imposes on the provinces, and the result is a diagnostic of how unusual the observed alignment is.

The wild cluster bootstrap addresses a different question (Cameron et al. 2008). It takes the province-level residual clusters as given, imposes the null and resamples their signs, so that it measures how often a coefficient of the observed size arises from the sampling variation within the 27 clusters when nothing is permuted. By this test the all-goods estimate is not significant at conventional levels, with p-values well above the randomisation values. The two procedures therefore agree on the sign and disagree on the strength of the evidence. A reader who treats the provinces as a sample from a larger population of provinces will prefer the bootstrap; a reader who asks whether the observed pairing of pressure and gradient change could be chance within these 27 provinces will prefer the permutation test. The paper reports both throughout, and its conclusions rest on the pattern across outcomes, specifications and placebos; no single p-value decides them.

B.4 Sensitivity Analysis

Table B2 changes one measurement choice at a time in the all-goods specification, without and with the controls, and reports the coefficient with both inference procedures. The binary pressure measure gives a larger coefficient with a wider interval, as the loss of variation from dichotomising a continuous moderator implies; its bootstrap p-value is the lowest in the table, because the comparison no longer leans on the scale of pressure at the extremes. A binary remoteness measure, road distance above the within-province median together with terrain relief above the national median (You et al. 2018), is the construction of the earlier version of this project and gives a smaller coefficient on its coarser scale with the same inference. Road distance computed on the 1 January 2016 network, which has gaps in several western provinces, reproduces the 2018 network almost exactly, so that the choice of snapshot is immaterial. The log construction gives zero for the all-goods index, a finding to which I return below.

Appendix B.6 reports two further checks on the composition of the sample. Weighting each county by its county-town population, so that the estimate describes the average resident of a county town, and dropping the county towns on the national list of 832 poverty counties both leave the gradient at or above its baseline size, and the event studies of both samples open the gap in 2019 (Table B5). The deep-poverty programme, whose designated areas overlap the remote county towns of several provinces, is examined in Section 7.3.

The log construction in the last pair of columns of Table B2 is a finding in its own right. The main construction measures each item per 10,000 residents at a fixed population, so that within-county changes are changes in the stock itself divided by a constant; the log construction measures proportional changes, in which a county that doubles a small stock counts as much as a county that doubles a large one. The gradient is thus a gap in how much the stock per resident grew, and it is largest where the stock per resident was already high. In proportional terms the goods as a whole did not grow at different rates across the remoteness gradient, and only the invisible network goods did (Appendix Table B3). Section 6.4 returns to this point.

Table B2: Robustness of the all-goods estimate to the measurement of the index, pressure and remoteness.

	Binary Discipline		Far half and rugged		2016 network		Log quantities	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post × Remote × Discipline	−0.277*	−0.235*	−0.078**	−0.062*	−0.186**	−0.152*	0.001	0.010
	(0.138)	(0.134)	(0.036)	(0.032)	(0.082)	(0.082)	(0.032)	(0.031)
Post × Remote	0.184	0.135	0.034	0.018	0.049	0.019	0.010	0.001
	(0.121)	(0.113)	(0.031)	(0.028)	(0.063)	(0.063)	(0.034)	(0.034)
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Province × Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls		✓		✓		✓		✓
Provinces	27	27	27	27	27	27	27	27
Counties	1,456	1,454	1,448	1,447	1,456	1,454	1,456	1,454
Observations	11,514	11,483	11,475	11,452	11,514	11,483	11,514	11,483
WCB p	0.076	0.108	0.218	0.279	0.157	0.281	0.969	0.758
RI p	0.034	0.071	0.012	0.025	0.003	0.026	0.977	0.773

Notes: Equation (6), all-goods index unless stated. Odd columns are without controls and even columns add the controls of Table 4, whose columns (1) and (2) report the main specification. (1)–(2) Discipline replaced by the indicator $D_p > 0.2$. (3)–(4) Remote replaced by the indicator for road distance above the within-province median and terrain relief above the national median. (5)–(6) Road distance on the 1 January 2016 OpenStreetMap snapshot. (7)–(8) Index built from items entering as log(1 + quantity). Standard errors clustered by province in parentheses; WCB and RI as in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5 Proportional Quantities and Binary Pressure

Table B3 reports the main specification with every quantity item entering as log(1 + quantity), so that the indices measure proportional change. The all-goods index under either weighting and the visible index show no gradient on this scale, and the visible index moves in the other direction; only the invisible index falls, as discussed in Appendices B.4 and B.7. Table B4 reports the main specification with the binary pressure measure, which splits provinces at the gap in the pressure distribution and underlies the two-group reading of Section 6.1.

Table B3: Proportional quantities: items enter as log(1 + quantity).

	All goods							
	Equal weights		Principal component		Visible		Invisible	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post × Remote × Discipline	0.001	0.010	−0.030	−0.020	0.061*	0.067**	−0.082*	−0.074*
	(0.032)	(0.031)	(0.031)	(0.030)	(0.030)	(0.030)	(0.044)	(0.043)
Post × Remote	0.010	0.001	0.015	0.005	0.011	0.003	0.006	−0.002
	(0.034)	(0.034)	(0.032)	(0.032)	(0.033)	(0.032)	(0.041)	(0.041)
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Province × Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls		✓		✓		✓		✓
Provinces	27	27	27	27	27	27	27	27
Counties	1,456	1,454	1,456	1,454	1,456	1,454	1,456	1,454
Observations	11,514	11,483	11,514	11,483	11,506	11,476	11,515	11,483
WCB p	0.969	0.758	0.572	0.638	0.088	0.073	0.343	0.380
RI p	0.977	0.773	0.335	0.503	0.088	0.048	0.046	0.073

Notes: As Table 4, with each quantity item entering as log(1 + quantity) and shares as shares, standardised on 2015–2017. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Discussion draft [do not cite or distribute] November 2026

Table B4: Binary pressure: provinces above and below the cut at $D_p = 0.2$.

	All goods							
	Equal weights		Principal component		Visible		Invisible	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post × Remote × Discipline	−0.277*	−0.235*	−0.341**	−0.300**	−0.157	−0.125	−0.322**	−0.291*
	(0.138)	(0.134)	(0.142)	(0.138)	(0.115)	(0.116)	(0.154)	(0.153)
Post × Remote	0.184	0.135	0.262**	0.212*	0.141	0.103	0.145	0.110
	(0.121)	(0.113)	(0.125)	(0.117)	(0.094)	(0.093)	(0.134)	(0.132)
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Province × Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls		✓		✓		✓		✓
Provinces	27	27	27	27	27	27	27	27
Counties	1,456	1,454	1,456	1,454	1,456	1,454	1,456	1,454
Observations	11,514	11,483	11,514	11,483	11,506	11,476	11,515	11,483
WCB p	0.076	0.108	0.027	0.057	0.204	0.331	0.069	0.116
RI p	0.034	0.071	0.009	0.020	0.166	0.267	0.036	0.045

Notes: As Table 4, with Discipline replaced by an indicator for $D_p > 0.2$ (sixteen provinces above, eleven below). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.6 Population Weights and the 832 Poverty Counties

Table B5 addresses the composition of the sample. Columns (1) and (2) weight each county by its 2015 to 2017 county-town population, so that the estimate describes the average resident of a county town. Without controls the coefficient is −0.18, 96 percent of the unweighted value; the bootstrap p-value falls to 0.05 and the randomisation p-value is unchanged at 0.003. Estimated separately by tercile of county-town population, the unweighted gradient is negative in the smallest, the middle and the largest third, and largest in the largest third, so that the result appears in county towns of every size. Columns (3) and (4) drop the 729 county towns on the national list of 832 poverty counties, the broader programme of which the deep-poverty areas are the core; 727 county towns in 25 provinces remain, and Tibet and Qinghai, whose county towns are all on the list, leave the sample. Without controls the coefficient is −0.23, larger than the baseline, with a clustered p-value of 0.08 and a randomisation p-value of 0.03. When the list enters as terms, $\mathsf{Post}_t \times \text{Poor}_i$ and $\mathsf{Post}_t \times \text{Poor}_i \times D_{p(i)}$, the poverty counties show a relative gain after 2018 that grows with pressure, +0.14 per standard deviation, which is the direction of the deep-poverty terms in Table 7, and the capital gradient is −0.23 with a randomisation p-value below 0.01. With controls, in columns (2) and (4), the two checks give −0.15 and −0.19.

B.7 Goods Families and Items

This check splits the all-goods index by family and by item. Under the pandering logic of Prat (2005) and Maskin and Tirole (2004), in which agents choose the actions a principal observes most easily, the conspicuous goods that a superior's visit can confirm would carry the gradient. The estimates of Table 4 run the other way. The invisible index moves more than the visible index in point estimate, by about twice as much without controls, and it is more precisely estimated. Because the two indices are standardised separately, the comparison requires a direct test. A regression of the difference between the invisible and the visible index, estimated with equation (6) on the county-years where both indices are observed, gives a negative coefficient on $\mathsf{Post}_t \times \text{Remote}_i \times \text{Discipline}_p$ that does not reach conventional significance under either inference procedure, with and without controls.[1] In per-capita

[1]The coefficient on the difference between the invisible and the visible index is −0.11 with a clustered standard error of 0.07; the bootstrap and randomisation p-values are 0.17 and 0.10, and the estimate with controls is the same. With the log construction the

Discussion draft [do not cite or distribute] November 2026

Table B5: Population weights and the 832 poverty counties.

	Population weights		832 poverty counties dropped	
	(1)	(2)	(3)	(4)
Post × Remote × Discipline	−0.179**	−0.151**	−0.228*	−0.192
	(0.076)	(0.072)	(0.123)	(0.121)
Post × Remote	−0.047	−0.069	0.002	−0.025
	(0.071)	(0.071)	(0.106)	(0.111)
County FE	✓	✓	✓	✓
Province × Year FE	✓	✓	✓	✓
Controls		✓		✓
County-town population weights	✓	✓		
832 poverty counties dropped			✓	✓
Provinces	27	27	25	25
Counties	1,456	1,454	727	726
Observations	11,514	11,483	5,763	5,746
WCB p	0.050	0.075	0.131	0.190
RI p	0.003	0.009	0.030	0.063

Notes: Equation (6), all-goods index. Odd columns are without controls and even columns add the controls of Table 4. (1)–(2) Weighted least squares with the county's 2015–2017 county-town population as weight; the wild cluster bootstrap and the permutation test run on the weighted model. (3)–(4) The 729 county towns on the national list of 832 poverty counties dropped. Standard errors clustered by province in parentheses; WCB and RI p-values from the original run of these checks, which uses its own seed. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

levels both families fall and the difference between them is not established. In proportional terms, with the log construction, the visible index rises and the invisible index falls, and the difference is precisely estimated under both procedures.

Figure B2 estimates equation (6) item by item, with the controls and each item standardised on 2015 to 2017, and shows where the family result comes from; its two panels are the two families of Figure 4. Among the goods scored by site visits, three items carry the gradient: the road surface, recorded as an area and the largest single estimate in the figure; street lamps; and public toilets. The other visible goods do not move, and two of them, the lit share of roads and park green area, move in the other direction. Among the invisible goods, drainage pipes, estimated precisely, and water-supply pipes, estimated less precisely, carry the gradient, together with garbage treatment capacity, a visible item recorded as a capacity figure; sewage pipes, sewage-plant capacity, water production capacity and gas pipes do not move. Without the controls the pattern is the same.

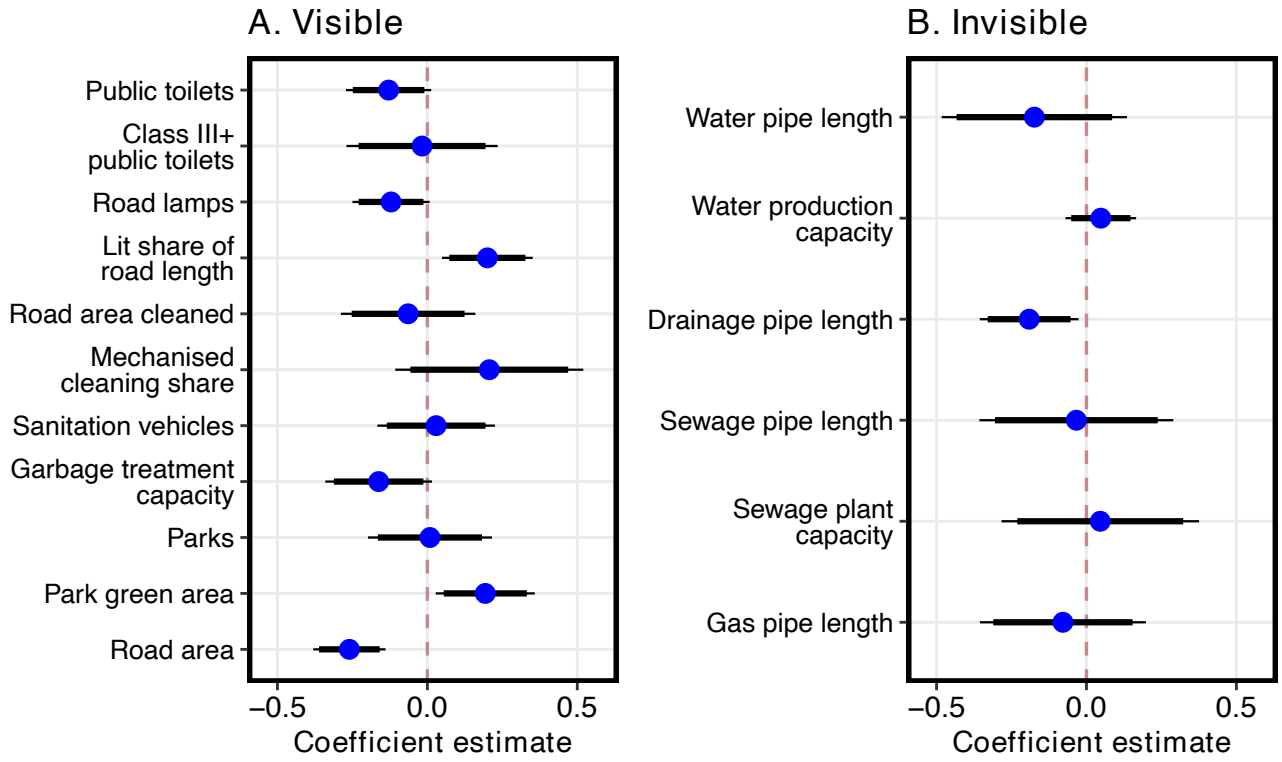
The goods that fall behind are those recorded as lengths and areas, the stock figures that a report carries and a visit confirms with difficulty: how many kilometres of pipe lie under a county town, or how many square metres of road surface it maintains, is known to the province from the county's return. The goods that hold up or rise are those a visit confirms at a glance: whether the roads are lit, how green the park is, how many vehicles the sanitation bureau runs. Public toilets are the exception that fits the same rule from the other side: they are the most counted facility of all, and under pressure the count fell behind in remote county towns because counting them is what the inspection tour does when it comes, and it comes less often. One reading of this pattern is that goods known to the province only through reports lose first once the campaign has taught evaluators to distrust reports. That reading was formed after the estimates were seen, and the direct test leaves the difference between the families open in per-capita levels; the item-level evidence narrows the readings without settling them. The split serves as a check on the all-goods estimate and does not carry the argument.

difference is −0.14 with a standard error of 0.04, and the bootstrap and randomisation p-values are 0.02 and below 0.01.

Discussion draft [do not cite or distribute] November 2026

Discussion draft [do not cite or distribute] November 2026

Figure B2: Item-level estimates of the pressure-specific gradient.



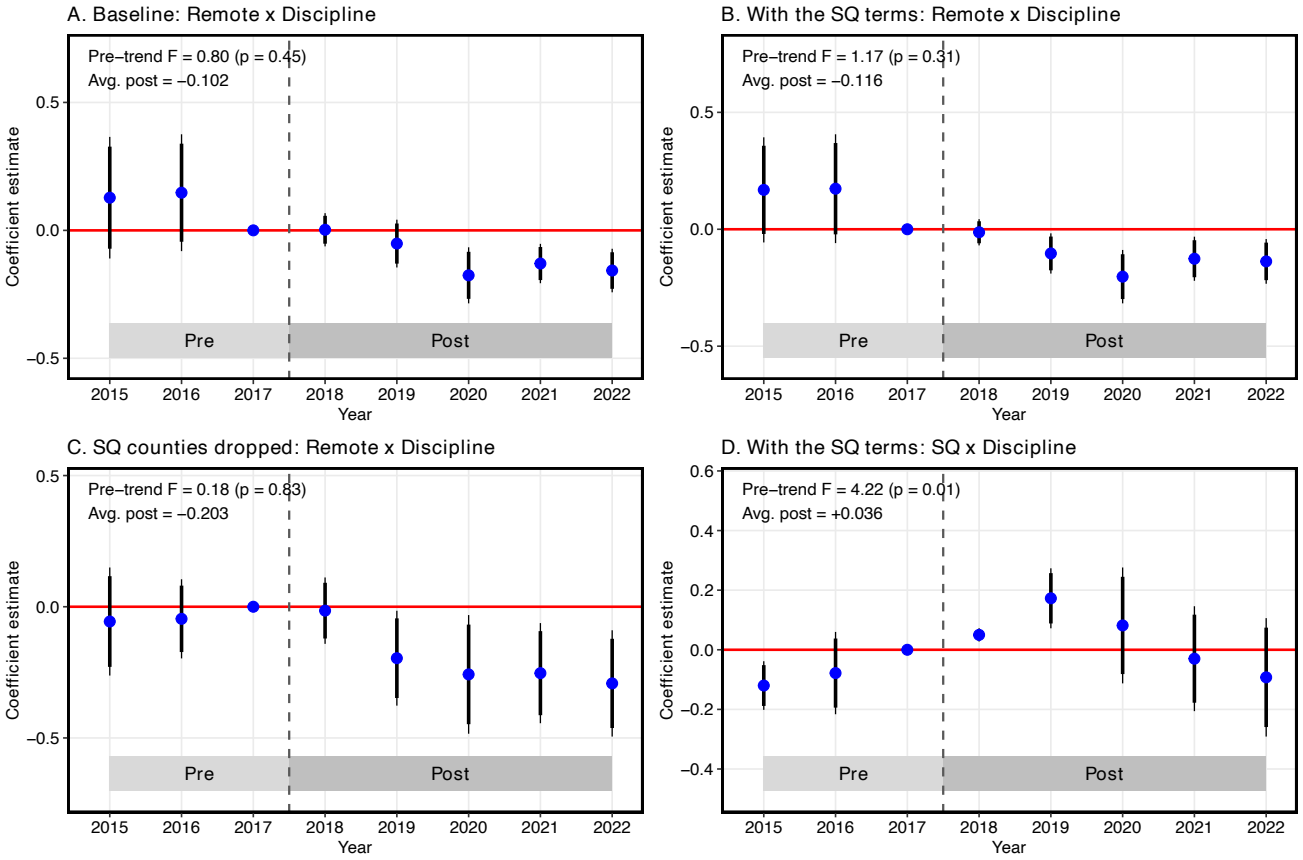
Notes: Coefficient on Post × Remote × Discipline from equation (6) for each of the seventeen items, standardised on 2015–2017, with county and province × year fixed effects and the controls of Table 4. Panel A: the eleven visible items; panel B: the six invisible items, as in Figure 4. Thick bars are 90% and thin bars 95% intervals from province-clustered standard errors. The largest estimates are road area (−0.26 standard deviations), drainage pipes (−0.19), water-supply pipes (−0.17), garbage treatment capacity (−0.16), public toilets (−0.13) and street lamps (−0.12); the lit share of roads (+0.20) and park green area (+0.19) move in the other direction. Without the controls the estimates are close to these.

B.8 Event Studies with the Deep-Poverty Areas Held Fixed

Figure B3 plots the event studies that accompany Table 7 of Section 7.3.

Discussion draft [do not cite or distribute] November 2026

Figure B3: Event studies with the deep-poverty areas held fixed.



Notes: All-goods index, no controls, reference year 2017; county and province × year fixed effects; thick bars 90% and thin bars 95% intervals from province-clustered standard errors. A: baseline, year × Remote × Discipline. B: the same coefficients with year × Deep poverty and year × Deep poverty × Discipline added. C: the designated county towns dropped. D: the year × Deep poverty × Discipline coefficients of panel B, the path of the deep-poverty push itself.

C Text Analysis

This appendix complements Section 3.3 with two tables and two figures: the regression estimates for the combined index under further fixed effects and trends, and for its interactions with pressure and remoteness; the share of county and prefecture reports that use each supervisory term; and the co-occurrence network of the goods and the accountability and verification terms in county reports before and after 2018.

Table C1 takes the combined index of Table 2, column (1), through the fixed effects and trends that the interrupted time series admits. Year fixed effects cannot enter it, because the break is common to every unit in a year, but fixed effects that vary within a year can. County reports are delivered at sessions of the local people's congress held between December of the preceding year and, in 2020, the early summer, and the first session after an election, which for most counties fell in the 2012, 2017 and 2022 reports, hears a review of the past five years. With county fixed effects alone the level shift is somewhat larger (column 1). The fixed effects for the month of the session change little (column 2), and those for the number of the session account for most of the difference (column 3, the specification of Table 2). Fixed effects for each county and congress term in place of the county and session-number fixed effects, the length of the report and county-specific linear trends leave the level shift close to that estimate, and the Driscoll–Kraay standard errors, which allow for shocks common to every unit in a year, keep it significant.

Discussion draft [do not cite or distribute] November 2026

The prefecture and provincial reports show the same break as the county reports (columns 7 and 8).

Table C2 turns to the comparisons that admit year fixed effects, in which the break is interacted with provincial pressure or with remoteness. In every column the time fixed effects absorb the break itself, and what remains identified is how it differed across provinces and county towns. Columns (1) to (4) ask whether the jump was larger where pre-2018 pressure had been higher. With year fixed effects, or with fixed effects for each region and year, the estimate is negative and imprecise; with province- or county-specific trends it is close to zero, so that the small negative difference reflects the provinces' paths before 2018. Columns (5) to (8) make for the language of the reports the comparison that the paper makes for public goods. After 2018, remote county towns in high-pressure provinces used the new vocabulary somewhat less than accessible ones in the same province and year. The estimate keeps its size when the gradient in remoteness may differ freely from year to year (column 6) and grows with county-specific trends while losing precision (column 7). The wild cluster bootstrap does not reject the null in any of these columns, and randomisation inference places the estimate at the margin of significance in columns (5) and (6). With prefecture-by-year fixed effects (column 8) the estimate turns positive and is too imprecise to inform, because little of the variation in remoteness lies within prefectures (Appendix D.1).

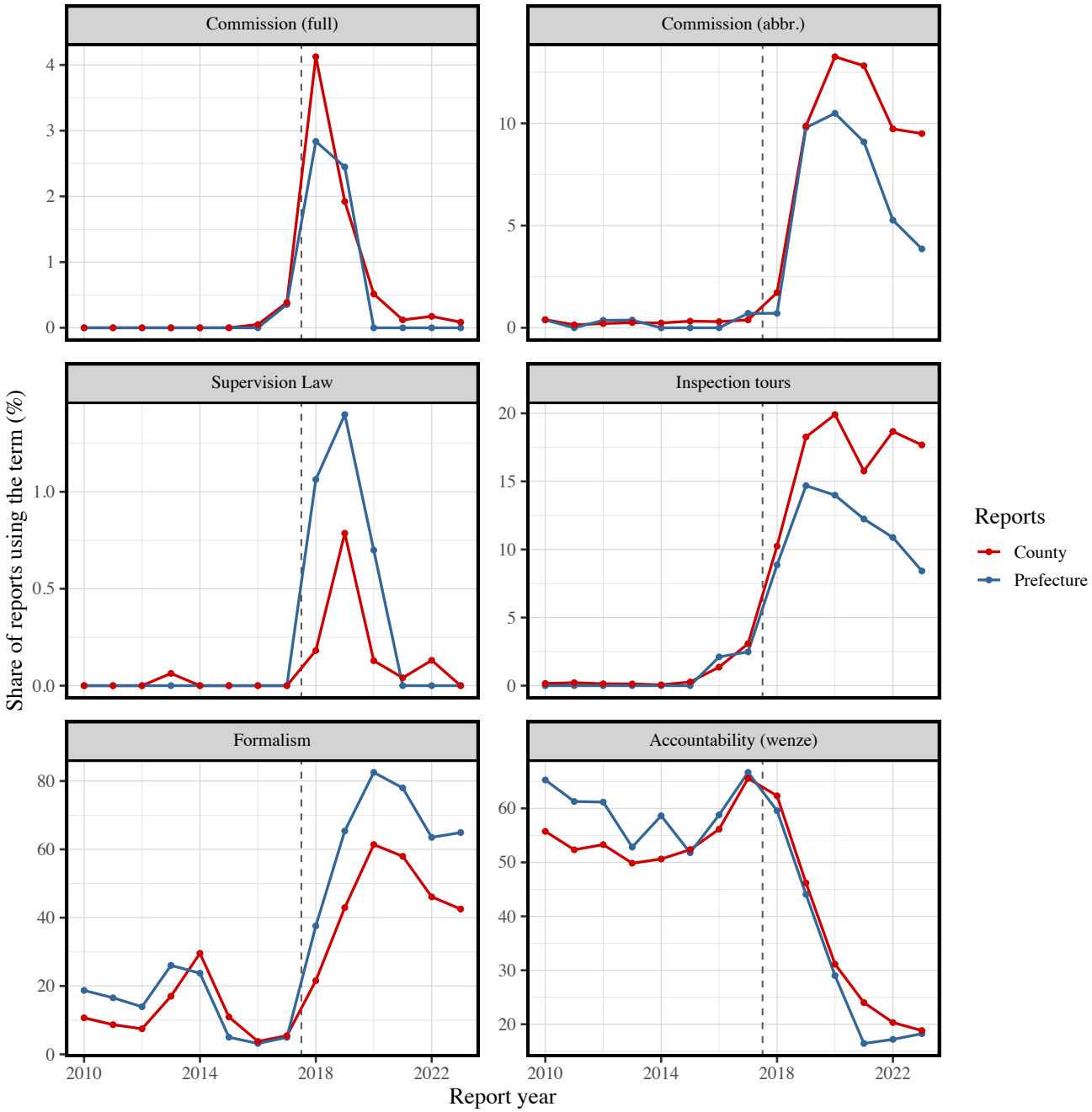
Table C1: Accountability language in government work reports: interrupted time series under further fixed effects and trends.

	Interrupted time series, county reports						Other tiers	
	County FE only (1)	Session month FE (2)	Session number FE (3)	County × term FE (4)	Report length (5)	County trends (6)	Prefecture reports (7)	Provincial reports (8)
Level shift at 2018 (β_2)	0.320***	0.316***	0.273***	0.255***	0.265***	0.257***	0.355***	0.451***
	(0.042)	(0.042)	(0.039)	(0.041)	(0.040)	(0.045)	(0.038)	(0.136)
Slope change after 2018 (β_3)	−0.099***	−0.098***	−0.095***	−0.054***	−0.094***	−0.093***	−0.066***	−0.099**
	(0.009)	(0.010)	(0.010)	(0.011)	(0.010)	(0.010)	(0.020)	(0.043)
Slope before 2018 (β_1)	0.012**	0.012**	0.017***	−0.001	0.016**		−0.004	0.010
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)		(0.011)	(0.022)
Log report length					0.115***	0.119***		
					(0.021)	(0.022)		
Unit FE	✓	✓	✓		✓	✓	✓	✓
County × term FE				✓				
Session-month FE		✓	✓	✓	✓	✓	✓	✓
Session-number FE			✓		✓	✓	✓	✓
County-specific linear trends						✓		
Reports	27,171	27,171	27,171	27,171	27,171	27,171	3,900	434
Units	2,699	2,699	2,699	2,699	2,699	2,699	286	31
Province clusters	31	31	31	31	31	31	31	31
Pre-2018 mean	0.692	0.692	0.692	0.692	0.692	0.692	0.707	0.872
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS
Wild cluster bootstrap p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001
Driscoll–Kraay p	0.005	0.004	0.012	0.002	0.015	0.018	0.015	0.016

Notes: Combined index of the five dictionary terms, mentions per 10,000 characters. Equation (1) on all reports of the tier with more than 200 characters, 2010–2023. Column (1) has county fixed effects alone. Column (2) adds fixed effects for the month of the session of the people's congress at which the report was delivered, and column (3) fixed effects for the number of that session within the congress's five-year term, which gives the specification of Table 2, column (1). Column (4) replaces the county and session-number fixed effects with fixed effects for each county and congress term, within which the session number moves with the year. Column (5) adds the log of the report's length in characters to column (3), and column (6) adds county-specific linear trends to column (5), which absorb the common trend. Columns (7) and (8) estimate the specification of column (3) on the prefecture and provincial reports, with prefecture or province fixed effects. The month, the number of the session and the congress term are read from the head of the report or from its title; reports that do not state them form a category of their own. Standard errors clustered by province in parentheses. The wild cluster bootstrap p-value (999 Rademacher draws, null imposed) and the Driscoll–Kraay p-value test the level shift. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Discussion draft [do not cite or distribute] November 2026

Figure C1: Share of reports using each term, county and prefecture reports.



Notes: Share of reports with at least one mention of the term. The supervisory commission appears in 2018 and is abbreviated from 2019; inspection tours step up in 2018 and stay; formalism peaks in the 2020 reports, which review 2019; the generic word for accountability (问责) declines after 2018.

Table C2: Pressure, remoteness and accountability language in county work reports under further fixed effects.

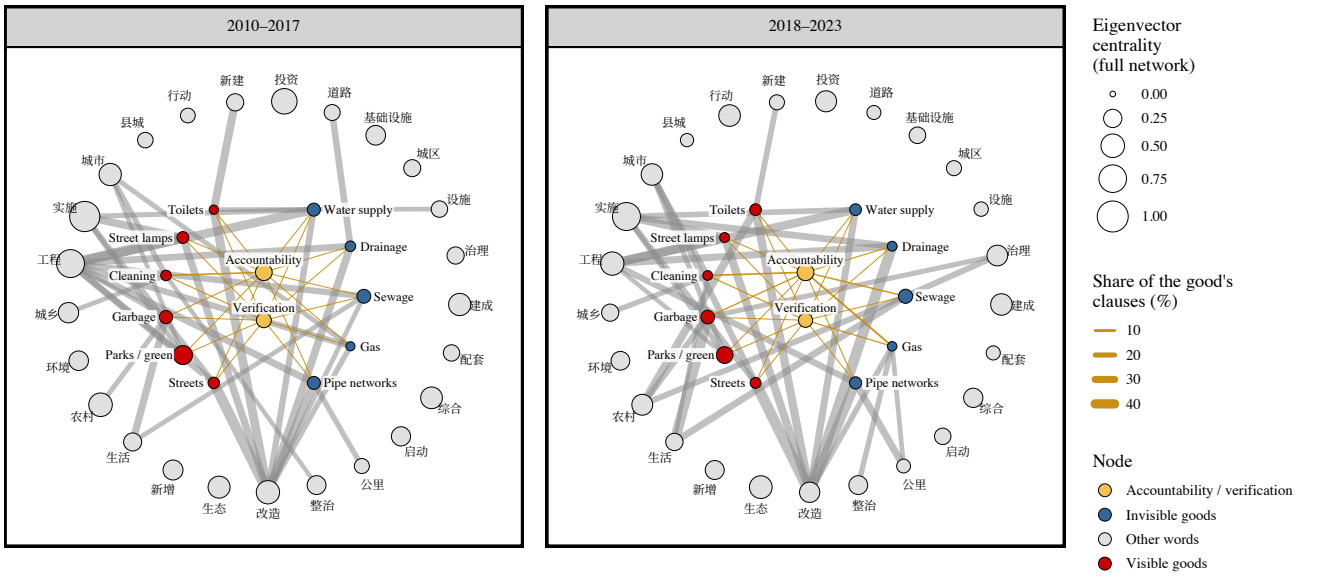
	Post × Discipline				Post × Remote × Discipline			
	Year FE	Region × year FE	Province trends	County trends	Province × year FE	Remote × year FE	County trends	Prefecture × year FE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post × Discipline	−0.064	−0.071*	−0.018	0.021				
	(0.042)	(0.041)	(0.046)	(0.049)				
Post × Remote					0.122		0.143	−0.024
					(0.075)		(0.101)	(0.205)
Post × Remote × Discipline					−0.136	−0.135	−0.177	0.152
					(0.080)	(0.081)	(0.132)	(0.200)
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓		✓	✓				
Region × Year FE		✓						
Province × Year FE					✓	✓	✓	
Prefecture × Year FE								✓
Remote × Year FE						✓		
Province-specific linear trends			✓					
County-specific linear trends				✓			✓	
County-years	13,279	13,279	13,279	13,279	13,279	13,279	13,279	13,279
Counties	1,370	1,370	1,370	1,370	1,370	1,370	1,370	1,370
Province clusters	27	27	27	27	27	27	27	27
Pre-2018 mean	0.705	0.705	0.705	0.705	0.705	0.705	0.705	0.705
Estimator	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS
Wild cluster bootstrap p	0.128	0.090	0.690	0.666	0.116	0.121	0.177	0.490
Randomisation inference p	0.166	0.176	0.716	0.669	0.058	0.061	0.163	0.415

Notes: Combined index of the five dictionary terms, mentions per 10,000 characters, in county reports without State Council texts and duplicated texts, averaged within county and year, for the county towns of the estimation sample of Table 4, 2010–2023. Post equals one from the 2018 reports, Discipline is as in Table 4 and Remote is the within-province percentile of road distance. Regions are the east, the centre, the west and the north-east of the National Bureau of Statistics. Remote × Year fixed effects are year-specific slopes in remoteness, which absorb Post × Remote. Prefectures are the prefecture-level units to which the county towns belong. Standard errors clustered by province in parentheses. The wild cluster bootstrap p-value (999 Rademacher draws, null imposed) and the randomisation inference p-value (999 permutations of Discipline across the 27 provinces) test Post × Discipline in columns (1) to (4) and Post × Remote × Discipline in columns (5) to (8). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Discussion draft [do not cite or distribute] November 2026

Discussion draft [do not cite or distribute] November 2026

Figure C2: Public goods and accountability language in county work reports: the co-occurrence network before and after 2018.



Notes: Clause-level co-occurrence in the county government work reports of 2010–2017 and 2018–2023. Nodes are the eleven goods categories (red: visible goods; blue: invisible goods), the accountability and verification term groups (gold) and the other words most connected to the goods (grey); node size is eigenvector centrality in the full clause network of the period, and the gold edges give the share of a good's clauses that carry the term group.

Discussion draft [do not cite or distribute] November 2026

D Further Evidence on the Mechanism

This appendix reports two further pieces of evidence on the mechanism: whether the gradient follows the provincial capital or the county's own prefecture seat, and whether the devices that certified performance before the reform depended on distance.

D.1 Where Verification Originates

Hypothesis 2 holds that the gradient belongs to the provincial capital because that is where the institutions that verify a county's delivery are based. The placebo distances of Appendix B.2 show that distance to the economic centres of the coast does not absorb the capital gradient. A county town also has a nearer principal, the prefecture, whose seat houses the inspection teams and the departments that handle the county's affairs from day to day. Placing the within-prefecture percentile of distance to the prefecture seat beside the capital measure, on the county towns whose prefecture has at least four county towns in the sample, leaves the capital gradient at its full size and gives the seat gradient the opposite sign: under pressure, county towns far from their own prefecture seat did relatively better.[2] The verification that matters for the gradient is the province's.

The same exercise shows the limit of the design. The within-province gradient is identified from comparisons across prefectures. Fewer than a fifth of the within-province variation in remoteness lies within prefectures, where the median prefecture holds four county towns, and a specification with prefecture-by-year fixed effects has a minimum detectable effect more than twice the size of the estimate.[3] Within a prefecture the data cannot tell whether the county town nearer to the capital did better; across the prefectures of a province they can, and it is across prefectures that the gradient of this paper runs.

D.2 Certification and Distance before the Reform

Section 2.3 rests the mechanism on a property of the two devices that carried performance upward before the reform: neither the growth tournament nor patronage depended on where within the province an official served. The panel assembled by Li and Manion (2023) allows this property to be examined one tier above the county. It follows the Party secretaries of 333 prefecture-level cities from 2008 to 2017 and records, for every year, whether the secretary was promoted, whether the secretary had been appointed under the provincial Party secretary then in office, which is the patronage tie of Jiang (2018) and Li and Manion (2023), the growth of the city's GDP standardised within its province and year, and the secretary's individual characteristics. I measure the remoteness of a prefecture, R_c, by the within-province percentile of the great-circle distance from the seat of its government to the provincial capital, so that the capital takes the value zero and the most distant prefecture of each province the value one; 326 of the 333 cities can be placed. For the years before the campaign (2008–2012) and for its first five years (2013–2017), I estimate

$$\begin{aligned}\mathsf{Promoted}_{ct} = \alpha_c + \lambda_t + \beta_1\,\mathsf{Tie}_{ct} + \beta_2\,(\mathsf{Tie}_{ct} \times R_c) \\ + \beta_3\,\mathsf{Growth}_{ct} + \beta_4\,(\mathsf{Growth}_{ct} \times R_c) + \mathbf{X}'_{ct}\gamma + \varepsilon_{ct},\end{aligned} \tag{D1}$$

[2] Capital gradient −0.22 (0.09) with the seat measure added and −0.22 (0.09) alone; seat gradient +0.11 (0.06) with the capital measure and +0.09 (0.06) alone; randomisation p-values 0.02 for the capital and 0.07 for the seat; 1,151 county towns in 24 provinces. The seat is the location of the prefecture government and the seat percentile is taken within the prefecture.

[3] Post × Remote × Discipline is +0.06 (0.17) with county and prefecture-by-year fixed effects, against −0.19 (0.08) with province-by-year fixed effects; 18 per cent of the within-province variance of the remoteness percentile lies within prefectures.

where α_c and λ_t are city and year fixed effects, the first of which absorbs R_c, and $\mathbf{X}_{ct}$ comprises the individual controls of Li and Manion (2023).[4] The marginal effect of a tie at remoteness r is $\beta_1 + \beta_2 r$, and that of relative growth is $\beta_3 + \beta_4 r$. Because a linear interaction can mask a gradient that bends or that rests on few observations at one end of the distribution, I also report the binning estimator of Hainmueller et al. (2019), which divides R_c at its terciles and lets the effect of the treatment differ across the three bins:

$$\begin{aligned}\mathsf{Promoted}_{ct} = \alpha_c + \lambda_t + \sum_{k\in\{L,M,H\}} G_{kc}\left[\mu_k\,\mathsf{Tie}_{ct} + \eta_k\,\mathsf{Tie}_{ct}\,(R_c - \bar{r}_k)\right] \\ + \beta_3\,\mathsf{Growth}_{ct} + \beta_4\,(\mathsf{Growth}_{ct}\times R_c) + \mathbf{X}'_{ct}\gamma + \varepsilon_{ct},\end{aligned} \tag{D2}$$

where G_{kc} indicates that R_c falls in tercile k and $\bar{r}_k$ is the median of R_c within it, so that μ_k is the marginal effect of a tie at $\bar{r}_k$; the indicators themselves are absorbed by the city fixed effects, and the binning estimate for growth exchanges the roles of the two treatments. Both equations are estimated separately for the two periods, with standard errors clustered by city as in Li and Manion (2023).

Figure D1 reports both estimators, and Table D1 the underlying regressions. Before the campaign, neither the premium of a tie nor the reward to relative growth varies with the remoteness of the prefecture beyond what sampling error allows (panels A and C). The linear slopes are small against their standard errors, and the binned estimates lie inside the band of the linear model in each tercile, so that the linear form does not conceal a gradient at either end of the distribution; the binned reward to growth is lowest in the most remote tercile, but its interval covers both zero and the linear fit.[5] After 2013 the premium of a tie disappears, as Li and Manion (2023) report, and it disappears across the whole range of remoteness: the estimates are negative near the capital and close to zero far from it, and the difference lies within sampling error. The reward to relative growth is close to zero throughout (panels B and D). Pooling the two periods, the change in either reward after 2013 does not vary with remoteness, and the secretaries of remote prefectures were promoted as often after 2013 as before, in provinces of high and of low disciplinary pressure alike.[6] The devices that certified performance before the reform therefore carried it from distant prefectures as readily as from near ones, which is the property on which Section 2.3 rests the argument.

Two limits qualify this reading. The estimates for the tie are imprecise, so the evidence excludes a steep gradient in the premium of a tie and leaves room for a moderate one. The panel also ends in 2017 and observes prefecture secretaries, so it describes the evaluation that preceded the reform one tier above the county; how the reform changed the evaluation of county leaders lies outside it.

Table D1 reports the regressions behind Figure D1. Columns (1) and (2) are the linear interaction model of equation (D1) for the two periods. Column (3) pools the two periods and interacts every term with an indicator for the years from 2013, so that its triple interactions measure whether the change in the premium of a tie or in the reward to growth varied with remoteness. Columns (4) and (5) ask whether the secretaries of remote prefectures were promoted less often after 2013, with city and province-by-year fixed effects, first alone and then interacted with disciplinary pressure.

[4] Gender, ethnic minority, college education, age and years in office with their squares, years of work experience, and earlier service in discipline inspection, courts and law enforcement, organisation, propaganda and the Communist Youth League. Years of Party membership, the remaining control of Li and Manion (2023), is observed for 92 per cent of the secretary-years and is left out.

[5] Tie × Remoteness is −0.058 (0.100) before 2013 and 0.069 (0.085) after; Relative growth × Remoteness is −0.033 (0.029) and 0.005 (0.031). Standard errors clustered by city; Table D1 adds province clustering and the wild cluster bootstrap. Against a yearly promotion rate of about 6 per cent before 2013, the minimum detectable slope of the tie is about 0.3.

[6] Change after 2013 by remoteness: 0.088 (0.088) for the tie and 0.014 (0.036) for growth; Remoteness × Post-2013 is −0.009 (0.035) with province-by-year fixed effects, and its interaction with Discipline is 0.005 (0.051), randomisation p 0.89 (Table D1, columns 3 to 5).

Discussion draft [do not cite or distribute] November 2026

Discussion draft [do not cite or distribute] November 2026

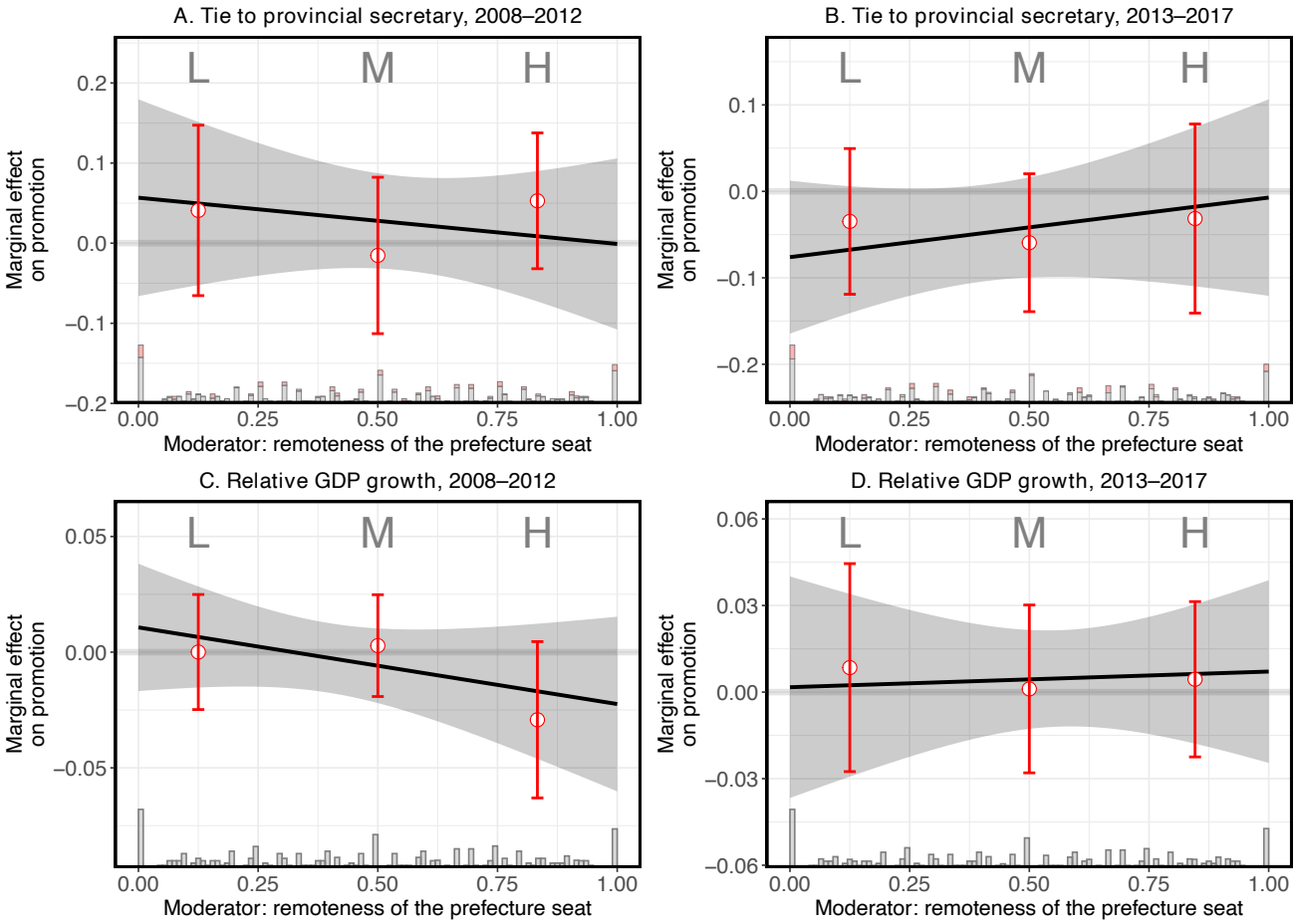
Table D1: Promotion of prefecture Party secretaries, certification devices and remoteness, 2008–2017.

	Returns by period		Change	Remote prefectures	
	(1)	(2)	(3)	(4)	(5)
Tie	0.057	−0.076*	0.041	0.014	0.014
	(0.062)	(0.045)	(0.044)	(0.019)	(0.019)
	[0.057]	[0.044]	[0.034]	[0.013]	[0.013]
Tie × Remoteness	−0.058	0.069	−0.049		
	(0.100)	(0.085)	(0.075)		
	[0.107]	[0.067]	[0.067]		
Relative growth	0.011	0.002	0.005	0.003	0.002
	(0.014)	(0.020)	(0.013)	(0.006)	(0.006)
	[0.014]	[0.019]	[0.015]	[0.006]	[0.006]
Relative growth × Remoteness	−0.033	0.005	−0.009		
	(0.029)	(0.031)	(0.024)		
	[0.030]	[0.029]	[0.028]		
Tie × Post-2013			−0.080		
			(0.054)		
			[0.047]		
Relative growth × Post-2013			−0.006		
			(0.019)		
			[0.019]		
Remoteness × Post-2013			−0.029	−0.009	−0.011
			(0.041)	(0.035)	(0.048)
			[0.034]	[0.030]	[0.048]
Tie × Remoteness × Post-2013			0.088		
			(0.088)		
			[0.081]		
Relative growth × Remoteness × Post-2013			0.014		
			(0.036)		
			[0.034]		
Remoteness × Post-2013 × Discipline					0.005
					(0.051)
					[0.052]
City FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓		
Province × Year FE				✓	✓
Individual controls	✓	✓	✓	✓	✓
Period	2008–2012	2013–2017	2008–2017	2008–2017	2008–2017
Observations	1,337	1,356	2,693	2,693	2,683
Cities	274	275	277	277	276
Provinces	25	27	27	27	26
Mean outcome	0.064	0.086	0.075	0.075	0.075
WCB p, tie	0.610	0.304	0.323		
WCB p, growth	0.302	0.858	0.690		
WCB p, remoteness				0.789	0.921
RI p					0.889

Notes: Linear probability models of the promotion of the prefecture Party secretary in the year, in the panel of Li and Manion (2023). Tie: appointment under the provincial Party secretary in office; Relative growth: city GDP growth standardised within province and year; Remoteness: within-province percentile of the great-circle distance from the prefecture seat to the provincial capital; Post-2013: the years from 2013; Discipline as in Table 4. Individual controls: gender, ethnic minority, college education, age and years in office with their squares, years of work experience, and earlier service in discipline inspection, courts and law enforcement, organisation, propaganda and the Youth League. Standard errors clustered by city in parentheses and by province in brackets; stars refer to the city-clustered standard errors. WCB p: wild cluster bootstrap with province clusters (999 draws, null imposed) for Tie × Remoteness and Relative growth × Remoteness in columns (1) and (2), for their products with Post-2013 in column (3), for Remoteness × Post-2013 in column (4) and for Remoteness × Post-2013 × Discipline in column (5); RI p permutes Discipline across provinces. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Discussion draft [do not cite or distribute] November 2026

Figure D1: Promotion returns to a patronage tie and to relative growth by the remoteness of the prefecture, 2008–2017.



Notes: Panels A and B: marginal effect of a tie to the provincial Party secretary (appointment under the secretary in office) on the probability that the prefecture Party secretary is promoted in the year; panels C and D: marginal effect of a one standard deviation rise in relative GDP growth. The moderator is the within-province percentile of the great-circle distance from the prefecture seat to the provincial capital. Black line and grey band: linear interaction model of equation (D1) with its 95% interval; red points and bars: binning estimator of equation (D2) at the medians of the remoteness terciles (L, M, H) with 95% intervals. Histograms: distribution of remoteness, in panels A and B split into secretary-years with a tie (red) and without (grey). City and year fixed effects and individual controls; standard errors clustered by city. 1,337 secretary-years in 274 cities in 2008–2012 and 1,356 in 275 cities in 2013–2017. Drawn with the interflex package of Hainmueller et al. (2019); the regressions are in Table D1.

Discussion draft [do not cite or distribute] November 2026

References

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427.

Hainmueller, J., Mummolo, J., and Xu, Y. (2019). How much should we trust estimates from multiplicative interaction models? Simple tools to improve empirical practice. *Political Analysis*, 27(2):163–192.

Ho, D. E. and Imai, K. (2006). Randomization inference with natural experiments: An analysis of ballot effects in the 2003 California recall election. *Journal of the American Statistical Association*, 101(475):888–900.

Jiang, J. (2018). Making bureaucracy work: Patronage networks, performance incentives, and economic development in China. *American Journal of Political Science*, 62(4):982–999.

Li, Z. and Manion, M. (2023). The decline of factions: The impact of a broad purge on political decision making in China. *British Journal of Political Science*, 53(3):815–834.

Maskin, E. and Tirole, J. (2004). The politician and the judge: Accountability in government. *American Economic Review*, 94(4):1034–1054.

Mastrorocco, N. and Teso, E. (2026). State capacity as an organizational problem: Evidence from the growth of the U.S. state over a century. *Review of Economic Studies*. Forthcoming.

Prat, A. (2005). The wrong kind of transparency. *American Economic Review*, 95(3):862–877.

You, Z., Feng, Z., and Yang, Y. (2018). Relief degree of land surface dataset of China (1 km). *Journal of Global Change Data & Discovery*, 2(2):151–155.